

Assignment 6: Mining and Proof-of-Work Lab

BLCH9X2 Blockchain |

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1. Purpose, and how to read this report alongside the notebook

This report accompanies the notebook **Assignment6_Bukho_Ndziweni_Blockchain_220081696**, submitted as HyperText Markup Language (HTML) with all outputs preserved, together with **mining.py**. Every claim below names the cell that produced it, using the execution numbers In [1] to In [6], and quotes the value that cell actually printed. The code was produced with the assistance of Artificial Intelligence and is commented thoroughly for clear understanding and interpretation of results.

Every timing figure in this report was measured on my own machine. Attempt counts, nonces and hashes are deterministic and reproduce anywhere, because timestamps are fixed constants and every nonce search starts at zero. Wall-clock seconds and hash rates are not: they were measured on the machine that ran the notebook, and are reported exactly as printed.

Sections 2 to 5 answer Part A. Section 6 answers Part B. Section 7 answers Part C.

Cell	What it does	Brief item
In [1]	Target arithmetic: difficulty, target and the rule this lab defines	A (a)
In [2]	The miner, and one block mined and verified	A (a)
In [3]	Parameter study: five difficulty levels, five trials each	A (b)
In [4]	Block reward paid to a miner address, and an attempt to redirect it	A (c)
In [5]	Retargeting from a deliberately wrong difficulty	B
In [6]	Retargeting after the hash rate is cut, and why this lab fixes difficulty	B

2. Part A (a): the rule this lab defines

A SHA-256 (Secure Hash Algorithm, 256-bit) hash is 64 hexadecimal characters, which is a very large whole number between 0 and 2 to the power 256, less one. Mining searches for a nonce whose hash, read as that number, falls at or below a ceiling called the target. Difficulty is how many times smaller than the maximum that ceiling is, so the target equals the maximum divided by the difficulty.

The rule stated, as the brief requires: a block is mined when its hash, read as a number, is at or below MAX_TARGET divided by the difficulty. This is the numeric target form rather than a count of leading zeros.

Two reasons for that choice, and both earned their keep later. Part B multiplies the target by fractional amounts during retargeting, which a count of zeros cannot express, since zeros move only in steps of sixteen. And the expected number of attempts falls out exactly as the difficulty itself, because a hash is equally likely to land anywhere in the range, so the chance of landing at or below the maximum divided by d is one in d. That gives section 3 a theoretical column to measure against rather than an approximation.

In [1] confirms the two forms are the same idea. At difficulty 65,536 the target is `0000ffff...`, so a hash must begin with four zeros, which is exactly the leading-zeros rule used in Assignment 5. The difference shows at difficulty 16,384, where the target is `0003ffff...`. Section 4 mined a block whose winning hash begins `0003cbbd`, a value that is valid under the numeric rule and inexpressible under a leading-zeros one.

3. Part A (b): parameter study of attempts and wall-clock time

Five difficulty levels were studied, each four times harder than the last, with five independent blocks mined at each. Each trial mines a different header, varied by block index, since re-mining identical content would repeat one nonce rather than produce five samples.

Difficulty	Expected attempts	Mean attempts	Median attempts	Mean seconds	Hashes per second
1,024	1,024	1,155	671	0.017	67,498
4,096	4,096	4,594	3,290	0.069	66,747
16,384	16,384	13,652	13,639	0.206	66,355
65,536	65,536	51,113	18,359	0.787	64,923
262,144	262,144	281,999	208,667	4.200	67,143

The whole study computed 1,762,564 hashes in 26.39 seconds, an overall rate of 66,777 hashes per second.

Mean time per block tracks difficulty closely. Each fourfold increase in difficulty produces roughly a fourfold increase in mean seconds, from 0.017 to 0.069 to 0.206 to 0.787 to 4.200. This is what should happen: the hash rate is a property of the processor, not of the difficulty, so time per block is simply expected attempts divided by a constant.

Attempt counts are far noisier than the means suggest. The five individual trials at difficulty 65,536 needed 1,313, 2,438, 39,395, 18,359 and 194,061 attempts, a spread of roughly 148 times within one difficulty level. That is why the median of 18,359 sits so far below the mean of 51,113. Mining attempts follow a geometric distribution, whose standard deviation is approximately equal to its mean, so five samples cannot pin the average down and the gap between the mean and expected columns should be read as sampling noise rather than as a defect in the rule.

The hash rate column is the internal consistency check. It ranges from 64,923 to 67,143 per second, a spread of 4 percent, which confirms that difficulty alone changed between levels. An earlier run of the same study produced a very different picture, rising from 46,569 to 68,262 per second, a spread of 47 percent. That run began on a cold processor: the easiest level completes in 0.12 seconds, before the processor leaves its idle clock speed, while the hardest runs for over 20 seconds under sustained load. The single block in In [2] supports the same reading, measuring 50,959 hashes per second from a cold start against 66,868 for the same difficulty moments later. Repeating the study on a warm machine removed the effect, which is reported here rather than discarded because it is a measurement artefact identified and eliminated rather than explained away.

4. Part A (c): the block reward

The reward is paid by a coinbase transaction placed first in the block. It has no sender, because the units it pays did not exist before the block was found, and it pays a fixed subsidy of 50.00 units plus the fees collected from the payments in that block, set at 0.25 units each. Three blocks were mined at difficulty 16,384 and credited to the address MINER-BK-01.

Block	Nonce	Attempts	Seconds	Fees	Reward	Hash (first 16)
1	6	7	0.000	0.50	50.50	0001a42b68b4232d
2	20,554	20,555	0.449	0.25	50.25	0003cbbdd53fc063
3	13,316	13,317	0.333	0.50	50.50	00003a34582836f4

The miner earned 151.25 units in total, of which 150.00 is newly created subsidy and 1.25 is fees. Sending addresses go negative in the resulting ledger, because no balance check is performed: this lab measures mining rather than transaction validity.

Block 1 was found in seven attempts. Against an expectation of 16,384 that looks wrong, and it is not. Expected attempts is an average over many blocks, not a quota, and any individual block can be found immediately. It is the same geometric distribution that produced the spread in section 3, seen from its lucky end.

The reward is protected by the proof of work rather than by a rule forbidding edits. Redirecting block 1's coinbase to another address after mining changed its hash from `0001a42b68b4232d... to bd743d194e737634...`, which no longer meets the target. A thief would have to redo the search, and having done so would be mining their own block rather than stealing this one.

5. Part A (d): proof of work against proof of stake for a regulated financial market infrastructure

A financial market infrastructure (FMI) is a payment system, central securities depository, securities settlement system, central counterparty or trade repository, supervised against the Principles for Financial Market Infrastructures issued by the Committee on Payments and Market Infrastructures (CPMI) and the International Organization of Securities Commissions (IOSCO). The question of which consensus mechanism suits such an institution is usually argued on energy. That is the least interesting of the three dimensions below.

5.1 Security

Proof of work (PoW) buys security outside the system. An attacker must acquire and run hardware, and the deterrent is the expenditure itself, incurred before any attack succeeds and repeated for as long as the attacker wants to stay ahead. The lab makes the shape of this concrete: at difficulty 262,144 a single block cost 281,999 hashes on average and 4.200 seconds on one thread. A network is that arithmetic multiplied, and the security budget is observable from outside as the aggregate hash rate.

Proof of stake (PoS) bonds security inside the system. A validator commits capital, and misbehaviour is punished by slashing, so the deterrent is confiscation after the fact rather than expenditure before it. Saleh (2021) gives the first formal economic model of PoS consensus and shows that a sufficiently modest reward schedule both supports a consensus equilibrium and precludes a persistent forking equilibrium. That result matters because it addresses the nothing-at-stake objection directly: validators are themselves stakeholders, so they internalise the effect of their behaviour on the value of the asset they hold, which a PoW miner selling into the market does not.

For a supervised institution the difference that bites is not which is stronger but which is measurable and which is stable. A PoW security budget is external and observable. A PoS security budget is denominated in the staked asset, so the cost of attacking the system falls exactly when that asset falls, which is to say in precisely the conditions where an FMI is most likely to be tested. That is a procyclical security budget, and a supervisor accustomed to countercyclical capital requirements should find it uncomfortable.

A second difference favours PoS. Supervision presumes identifiable, accountable operators. Anonymous miners cannot be licensed, examined or sanctioned, whereas a stake-based validator set can be permissioned and tied to legal identity without changing the consensus rule at all. PoS admits a whitelist; PoW does not, because its entire purpose is to let strangers participate.

5.2 Energy

PoW's energy consumption is not a side effect to be optimised away. It is the mechanism. The expenditure is what an attacker must match, so removing it removes the security. This lab spent 1,762,564 hashes and 26.39 seconds to produce blocks of no value whatever, and that expenditure was the product rather than a byproduct of it.

The magnitude of the alternative is now measured rather than argued. Ethereum's transition to PoS on 15 September 2022 reduced the network's energy consumption by approximately 99.95 percent according to the Ethereum Foundation, and by more than 99.988 percent according to the Crypto Carbon Ratings Institute, from roughly 23 million megawatt hours per year to just over 2,600. Whatever the exact figure, the difference is three orders of magnitude and not a matter of efficiency gains at the margin.

The argument that matters for an FMI, though, is not environmental. It is that PoW's energy bill purchases sybil resistance in an open market, and an FMI has no use for that property. Its participants are licensed institutions, known in advance and admitted by contract and regulation. Paying for permissionless entry, and paying for it continuously, is paying for something the institution is in fact obliged not to want.

5.3 Settlement finality

Principle 8 of the PFMI requires an FMI to provide clear and certain final settlement, at a minimum by the end of the value date, and intraday or in real time where necessary or preferable. Final settlement means an irrevocable and unconditional transfer or discharge of an obligation. The operative words are clear, certain and irrevocable.

PoW cannot supply them. Nakamoto (2008) is explicit that finality is probabilistic: the chance that an attacker rewrites history falls as blocks accumulate, and section 11 of the whitepaper computes that decay, but it never reaches zero. Assignment 5 measured the same property from the other side, pricing a two-block rewrite rather than forbidding it. A settlement rule that says reversal grows steadily more expensive is not the same kind of statement as one that says settlement is irrevocable at a stated moment. It is a difference of category, not of degree, and no number of confirmations closes it.

PoS designs with an explicit finality gadget come closer. Ethereum finalises checkpoints roughly every two epochs, about 12.8 minutes, and reversing a finalised checkpoint requires at least one third of all staked capital to be destroyed. That is still economic rather than absolute, but it names a moment and attaches a known penalty, which is a far easier object to write into a rulebook than a decaying probability.

Neither mechanism confers legal finality by itself. Legal finality comes from designation in law, and the protocol can at best supply a moment for the law to point at. That is the real test: does the design produce an identifiable instant, attributable to identifiable parties, that a legal regime can declare irrevocable? PoS with a finality gadget does. Open PoW does not.

5.4 Conclusion

For a regulated FMI, PoS is the better of the two on all three dimensions, but the honest conclusion is that the comparison is the wrong one. Both mechanisms exist to solve consensus among parties whose identities are unknown, and an FMI knows exactly who its participants are. Once the validator set is definable, a permissioned Byzantine fault tolerant design of the kind set out by Castro and Liskov (1999) delivers deterministic finality in a single agreement round, at a latency suited to real time gross settlement (RTGS), without buying open participation at all. The relevant question for an institution such as the South African Reserve Bank's settlement system is therefore not PoW against PoS, but what a distributed ledger of any kind adds over a well-run RTGS, and the answer has to be found in participant experience and resilience rather than in the consensus mechanism.

6. Part B: difficulty retargeting, and why this lab fixes difficulty

Bitcoin measures how long the last 2,016 blocks took and multiplies the target by the ratio of actual to expected time, clamped to a factor of four in either direction. In [5] applies the same arithmetic at

laboratory scale, with epochs of 5 blocks and a target block time of 0.20 seconds, starting from a difficulty set far too low on purpose.

Epoch	Difficulty	Seconds per block	Raw ratio	Next difficulty
1	256	0.0034	58.89	1,024
2	1,024	0.0128	15.62	4,096
3	4,096	0.0562	3.56	14,572
4	14,572	0.0686	2.92	42,489
5	42,489	0.8180	0.24	10,622
6	10,622	0.0877	2.28	24,234
7	24,234	0.3810	0.52	12,721
8	12,721	0.2327	0.86	10,933

The mean hash rate across the run was 56,562 per second, so the difficulty that fits this machine is 11,312. The algorithm finished at 10,933, within 3 percent of it.

Epochs 1 and 2 show the clamp binding upwards. The raw ratio demanded a 58-fold increase, and the algorithm refused, multiplying by four instead. Epoch 5 shows the same limit on the way down: a block time of 0.8180 seconds produced a ratio of 0.24, below the one-quarter floor, so the difficulty fell to 10,622 rather than to the 10,203 the raw ratio implied. The clamp is what stops one freak epoch, or a miner lying about a timestamp, from moving the difficulty somewhere absurd in a single step.

After convergence the difficulty oscillates rather than settling. It moves between 10,622 and 24,234 around a correct value of 11,312. This is the small sample problem from section 3 arriving in a new place: five blocks is far too few to measure a block time, so the algorithm is partly chasing noise. Bitcoin averages over 2,016 blocks, roughly two weeks of evidence, for precisely this reason.

6.1 Retargeting when the hash rate changes

In [6] cuts the effective hash rate part way through the run. The handicap is real rather than simulated: each attempt performs an extra throwaway hash costing the same as a genuine one, so the processor truly spends longer per attempt. Epochs 1 to 3 run at full speed and epochs 4 to 8 run slowed.

Epoch	Speed	Difficulty	Seconds per block	Hashes per second	Next difficulty
1	full	11,312	0.1478	52,913	15,310
2	full	15,310	0.2899	55,316	10,562
3	full	10,562	0.1465	55,995	14,419
4	halved	14,419	0.2047	28,842	14,091
5	halved	14,091	0.3781	28,086	7,452
6	halved	7,452	0.3863	29,447	3,858
7	halved	3,858	0.1211	29,349	6,374
8	halved	6,374	0.2002	28,821	6,368

The measured rate fell from 54,741 to 28,909 hashes per second, a reduction of 47 percent rather than the intended 50, which is the difference between a measurement and an assumption. Mean block time was 0.1947 seconds before the change and 0.2581 after, against a target of 0.2000, so the algorithm held its aim while the ground moved. The difficulty settled near 6,368, roughly half its starting value, which is the correct response to half the power.

The conclusion is worth stating carefully. Losing half the mining power does not halve the speed of the chain and does not halve its usefulness. It costs one slow epoch, after which block times recover at

a lower difficulty. What changes permanently is the security budget, because half as much work now stands behind each block, which is the same quantity the essay in section 5 identifies as the thing a supervisor would have to monitor.

6.2 Why this lab uses a fixed difficulty

Reproducibility comes first. With a fixed difficulty and fixed timestamps, every nonce and hash in the notebook is identical on any machine. Retargeting makes the difficulty a function of local processor speed, so nobody else could reproduce the values, only the shape of the behaviour, as the difference between my two runs of section 6 already shows.

The parameter study needs one variable at a time. Section 3 measures how attempts and time respond to difficulty, and if the difficulty moved during the run nothing could be attributed to anything.

There is also no network here and no adversarial clock. Bitcoin retargets on timestamps reported by miners who may lie, which is why both the clamp and the median-of-eleven timestamp rule exist. A single process with fixed timestamps has none of that context, so section 6 demonstrates the arithmetic of retargeting rather than the adversarial problem it was designed to solve.

Finally, five-block epochs are too short to measure anything reliably, as the oscillation above shows. Fixing the difficulty avoids reporting noise as though it were a result.

7. Part C: open questions for a regulated FMI

The essay in section 5 concludes that PoS suits a regulated FMI better than PoW on security, energy and finality, and that a permissioned design suits it better than either. Each of those conclusions rests on assumptions that a supervisor would have to test rather than accept. The following are open, in the sense that the literature cited here does not settle them.

1. At what moment does protocol finality become legal finality? Castro and Liskov (1999) deliver a commit that is deterministic under their fault bound, and Ethereum's finality gadget names a checkpoint, but legal irrevocability is conferred by designation in law rather than by a protocol. Can a probabilistic rule of the kind Nakamoto (2008) describes ever be designated, and if not, is PoW categorically excluded from FMI use rather than merely ill-suited?
2. How should a supervisor stress-test a security budget denominated in the asset being settled? If the cost of attacking a PoS system falls with the price of the staked token, the protection is procyclical. What is the PoS analogue of a capital requirement, and what would a supervisory stress scenario for it look like?
3. Is automated slashing compatible with administrative due process? Confiscating the capital of a licensed institution by algorithm, without hearing, sits awkwardly with the procedural protections such institutions normally enjoy. Who adjudicates an erroneous slash caused by an operational fault rather than misconduct?
4. What metric should govern validator concentration? Saleh (2021) models validators as stakeholders, but custodial staking concentrates voting power in a small number of service providers, which recreates the concentration that decentralisation was meant to avoid. What threshold should trigger supervisory action, and measured against what denominator?
5. How should energy be disclosed? Per transaction is the common metric and arguably the least meaningful, since it varies with block occupancy rather than with anything about the mechanism. Energy per unit of value settled, or per unit of security purchased, would be more informative and neither is standard.

6. How does an FMI running on a public chain execute an orderly recovery or wind-down of infrastructure it does not control? The PFMI expect a credible plan, and a plan that depends on the cooperation of an anonymous validator set is difficult to call credible.
7. Finally, and most awkwardly for the whole exercise: if the conclusion is a permissioned ledger with known validators, what measurable advantage does it retain over a modern RTGS? The consensus literature answers how to agree among strangers, and an FMI has no strangers.

8. References

Antonopoulos, A. M. and Harding, D. A. (2023) Mastering Bitcoin: Programming the Open Blockchain. 3rd edn. Sebastopol: O'Reilly Media. Mining and difficulty retargeting are covered in the mining chapter.

Castro, M. and Liskov, B. (1999) 'Practical Byzantine fault tolerance', Proceedings of the Third Symposium on Operating Systems Design and Implementation. New Orleans, February 1999.

CPSS-IOSCO (2012) Principles for Financial Market Infrastructures. Basel: Bank for International Settlements. Principle 8, settlement finality.

Ethereum Foundation (2022) The Merge. Available at ethereum.org. Energy consumption reduction of approximately 99.95 percent, with the Crypto Carbon Ratings Institute measurement of over 99.988 percent reported separately.

Nakamoto, S. (2008) Bitcoin: A Peer-to-Peer Electronic Cash System. Section 11, calculations on the probability of an attacker catching up.

Saleh, F. (2021) 'Blockchain without waste: proof-of-stake', The Review of Financial Studies, 34(3), pp. 1156-1190.

van Flymen, D. (2017) 'Learn Blockchains by Building One'. Chain construction and verification.

9. Code repository

<https://github.com/bukhondziweni19/blockchain-assignment-6>